Turtle Graphics: Polygons and a House

AP Computer Science A | Java

Name: _____ Period: _____ Date: _____

Overview

You will use the `Turtle` class to draw four polygons, then combine what you have learned to draw a house. You will practice loops, methods with parameters, and breaking a big problem into smaller pieces.

Turtle Methods You Must Use

Method	What it does
`penColor(Color c)`	Sets the color of the line the turtle draws
`width(int w)`	Sets the thickness of the line
`forward(int d)`	Moves the turtle forward d pixels
`backward(int d)`	Moves the turtle backward d pixels
`left(int degrees)`	Turns the turtle left
`right(int degrees)`	Turns the turtle right
`up()`	Lifts the pen (moving does not draw)
`down()`	Lowers the pen (moving draws)

Note to teacher: adjust parameter types or names to match your version of the Turtle class.

Part 1: Four Polygons

Draw **four different polygons**, each in a different color and line width. Examples: triangle, square, pentagon, hexagon, octagon.

Requirements

1. Each polygon must be drawn by a method such as `drawPolygon(Turtle t, int sides, int sideLength)`. Better yet, write one general method and call it four times.
2. Use a `for` loop inside the method. Do not write `forward` and `left` repeatedly by hand.
3. Calculate the turning angle with the formula `360 / sides`.
4. Before drawing each polygon, call `penColor` and `width`.
5. Use `up()` and `down()` to move the turtle to a new starting spot so the polygons do not overlap.
6. Use `backward` at least once (for example, to reposition the turtle).

Think about it: Why does `360 / sides` work? What happens with `sides = 7`, since 360 does not divide evenly? (Look at integer division.)

Part 2: Draw a House

Using your polygon methods or new methods, draw a house on the screen.

Required parts

- **Body:** a square or rectangle
- **Roof:** a triangle sitting on top of the body
- **Door:** a rectangle in a different color
- **At least one window:** a square in a different color
- **One more feature of your choice:** chimney, sun, tree, doorknob, path, etc.

Requirements

1. Each part of the house must be its own method (`drawBody`, `drawRoof`, `drawDoor`, and so on), called from `main` or a `drawHouse` method.
2. Use at least **three different colors** and **two different line widths**.
3. Use `up()` and `down()` to move between parts without drawing stray lines.
4. Use at least one method that takes parameters (size, position, or color).

Starter Code

```
public class TurtleHouse {

    public static void drawPolygon(Turtle t, int sides, int sideLength) {
        // TODO: use a loop with forward and left
    }

    public static void drawBody(Turtle t, int size) {
        // TODO
    }

    public static void drawRoof(Turtle t, int size) {
        // TODO
    }

    public static void drawDoor(Turtle t, int size) {
        // TODO
    }

    public static void drawHouse(Turtle t, int size) {
        // TODO: call your part methods, moving with up() and down()
    }

    public static void main(String[] args) {
        Turtle t = new Turtle();

        // Part 1: four polygons
        // TODO

        // Part 2: the house
        // TODO
    }
}
```

Planning Tips

- Sketch your house on graph paper first and label the lengths and angles.
- The roof's turning angle is different from the square's. An equilateral triangle turns **120°**.
- Test each method on its own before combining them.
- Leave the turtle facing the same direction it started in, so later methods are easier to write.

Grading Rubric (50 points)

Category	Points
Four distinct polygons drawn correctly	10
Loops used (no repeated hand-written code)	5
Methods with parameters used appropriately	8
`penColor` and `width` used meaningfully	4
`up`/`down` used to move without drawing	4
`backward` used at least once	2
House has all required parts, reasonable layout	12
Code style: indentation, naming, comments	5
Total	**50**

Extension Challenges

- Draw a **neighborhood** of three houses with different sizes and colors by calling `drawHouse` with different parameters.
- Add a street, sun, and clouds.
- Use a loop to draw a row of houses.
- Use `Math.random()` to pick random colors for each house.

Reflection Questions (submit with your code)

1. How did using methods make drawing the house easier than writing everything in `main`?
2. What would you have to change in your code to make the house twice as large?
3. What bug gave you the most trouble, and how did you fix it?